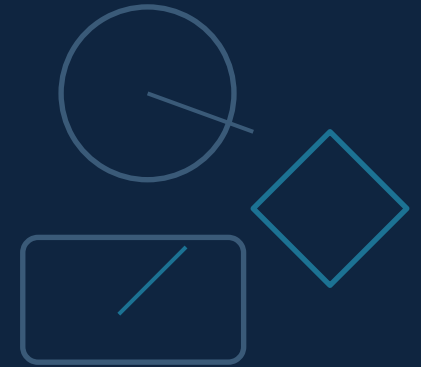


DISTRIBUTED SYSTEMS • PROCESS ORCHESTRATION

BPMN

Business Process Model and Notation

What it is, what it is used for, and the features that let it orchestrate work across distributed, service-based architectures.



What We'll Cover

01

What is BPM?

The discipline behind managing business processes

02

What is BPMN?

The standard notation for modeling those processes

03

Why It Matters

Purpose: a shared language for business and IT

04

Core Elements

Events, activities, gateways, lanes and artifacts

05

A Worked Example

Reading a real, cross-service BPMN diagram

06

BPMN in Distributed Systems

Engines that execute models as live workflows

07

Orchestration vs. Choreography

Two coordination styles for services

08

Engines, Benefits & Takeaways

Tools, adoption and why it scales

What Is BPM?

Business Process Management (BPM) is the discipline of identifying, modeling, executing, monitoring and continuously improving how an organization gets work done — its end-to-end business processes.

- **Model** — Capture how a process actually works, step by step.
- **Execute** — Run the process consistently, by people and by systems.
- **Monitor & Improve** — Measure performance and refine the process over time.



BPM is the discipline.

BPMN is the notation used to practice it.

What Is BPMN?



2005

Maintained by the Object Management Group (OMG) as a global standard

BPMN (Business Process Model and Notation) is a standardized set of graphical symbols for documenting business processes — precise enough that both people and software can read the same diagram the same way.



A visual model

Boxes, diamonds and arrows anyone can follow — no technical training required.



An executable model

Stored as XML, so a workflow engine can run the diagram directly, not just illustrate it.

Why BPMN Matters

It gives every stakeholder — analyst, developer, operator — one unambiguous picture of how work moves through a system.



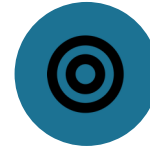
Common Visual Language

Removes the ambiguity of text-only process descriptions.



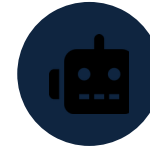
Bridges Business & IT

The same diagram analysts design is the one engineers implement.



Faster Alignment

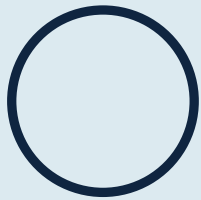
Teams agree on how a process works before it's built.



Enables Automation

Executable models can run directly on a workflow engine.

Core Elements: Flow Objects



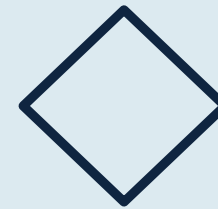
Events

A circle. Something that happens — starts, interrupts or ends a process (e.g. an order arrives, a timer fires).



Activities

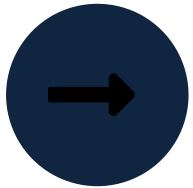
A rounded rectangle. A unit of work — a task performed by a person, a system, or a service.



Gateways

A diamond. A decision point that splits or merges the flow — exclusive, parallel or inclusive.

Core Elements: Connectors & Context



Connecting Objects

Sequence flows (solid) order steps within one participant; message flows (dashed) show communication between participants.



Swimlanes

Pools represent participants (e.g. two services); lanes divide a pool by role, making ownership explicit.

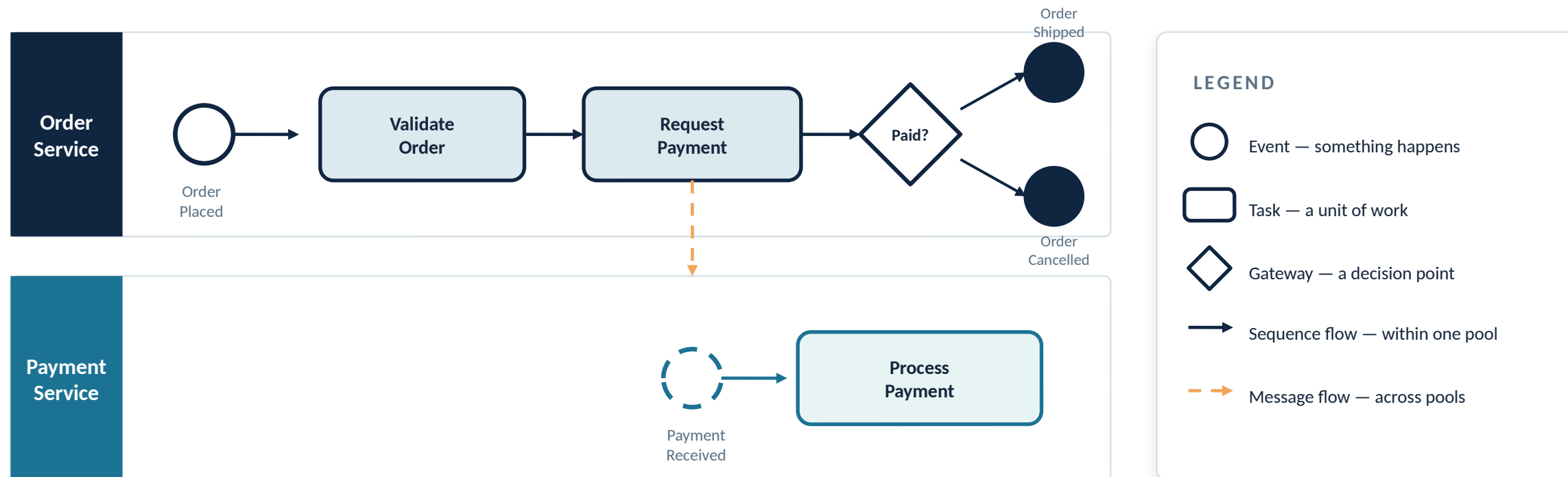


Artifacts

Data objects, groups and text annotations add supporting detail without changing the flow itself.

Reading a Cross-Service Process

Two pools, two independently owned services, coordinated by message flow — the same pattern a workflow engine automates.



Message flow (amber, dashed): the sequence flow inside each pool stays private to that service; only the message flow crosses the boundary — the same loose coupling microservices rely on.

BPMN in Distributed Systems

A BPMN model isn't just a picture — it is stored as XML with defined execution semantics. A **workflow engine** loads that model and runs it directly, one instance per case.



Coordinates independent services

Each task hands work to a separate microservice, which acts as a worker — the engine directs, it doesn't own the logic.



Maintains process state

The engine tracks exactly where each running instance is, surviving restarts, delays and long-running steps.



Keeps services loosely coupled

Services don't call each other directly or know the whole process — only the engine sees the full picture.

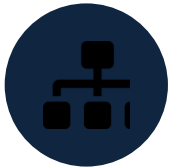
“

Flow requirements are timeless — the engine underneath is what scales.

Millions of workflow instances per second are achievable on modern BPMN engines (e.g. Zeebe) without changing the flow language.

Orchestration vs. Choreography

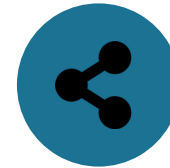
BPMN is most associated with orchestration, but both patterns coordinate the same distributed services.



Orchestration

A central engine directs

- One BPMN process model owns the sequence of steps
- Services act as workers, called by the engine
- Easy to see, trace and change the whole process
- Best when one team owns end-to-end control



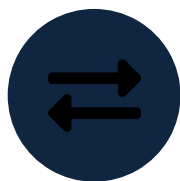
Choreography

Services react to events

- Each service knows only its own reaction to events
- No central coordinator — pure peer-to-peer messaging
- More autonomy per team, less global visibility
- Best for loosely governed, highly independent teams

Features That Make It Fit

BPMN 2.0 includes purpose-built constructs for the realities of distributed, asynchronous systems.



Message Correlation

Matches an incoming message to the exact running process instance it belongs to.



Timers & Timeouts

Combines receive tasks with timer events to handle slow or missing responses.



Error Boundary Events

Attaches predictable failure handling directly to the activity that can fail.



Multi-Instance & Parallel

Runs one step many times over a collection, or several branches at once.

Engines & Real-World Adoption

Open-source and commercial BPMN engines run these models as live, scalable workflows in production systems.



Camunda / Zeebe

Cloud-native BPMN engine built for microservice orchestration at high throughput.



Flowable / jBPM

Java-based open-source engines embedding BPMN execution into applications.



IBM & Appian BPM

Enterprise low-code platforms using BPMN as their process modeling core.

1M+/sec

new workflow instances modern engines like Zeebe are built to handle

2005

the year OMG standardized BPMN, still the dominant process notation

1

single model that spans design, automation and monitoring

Benefits & Key Takeaways

- ✓ One notation, understood by business and engineering alike
- ✓ Supports both orchestration and choreography-style designs
- ✓ Diagrams double as executable, automatable process definitions
- ✓ Backed by a mature standard (OMG) and a broad engine ecosystem
- ✓ Purpose-built for coordinating independent, distributed services
- ✓ Scales from a single workflow to millions of running instances

Bottom line: BPMN turns a business process into a diagram that both explains and runs the system — making it a natural fit for orchestrating work across distributed architectures.

Sources

- **IBM — What is Business Process Modeling and Notation (BPMN)?**
ibm.com/think/topics/bpmn
- **Object Management Group — Business Process Model & Notation (BPMN)**
omg.org/bpmn
- **Visual Paradigm — What is BPMN?**
visual-paradigm.com/guide/bpmn/what-is-bpmn
- **Camunda — BPMN for Microservices Orchestration, Part 1**
camunda.com/blog/2018/08/bpmn-for-microservices-orchestration-a-primer-part-1
- **Camunda — BPMN and Microservices Orchestration, Part 2**
camunda.com/blog/2018/08/bpmn-microservices-orchestration-part-2-graphical-models
- **Camunda — Microservices Orchestration Software**
camunda.com/platform/use-cases/orchestrate-microservices
- **TechTarget — What Is Business Process Modeling Notation (BPMN)?**
techtarget.com/searchcio/definition/Business-Process-Modeling-Notation